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Education

- 2024 📖 **PhD degree, Peking University** in Space Physics
- 2022 – 2024 📖 **Visiting PhD student, IRAP, Toulouse, France** in Space Physics
- 2019 📖 **Bachelor's degree, Peking University** in Space Physics

Fellowships, Awards and Scholarships

Fellowships

- 2025 📖 **Alexander von Humboldt Research Fellowship**, Alexander von Humboldt-Stiftung

Awards

- 2024 📖 **Best Student Paper Awards**, National Planetary Science Conference.
- 2023 📖 **Merit Student**, Peking University.
- 2022 📖 **Best Student Paper Awards**, Chinese Geoscience Union (CGU).
📖 **Best Student Poster Awards**, Asia Oceania Geosciences Society (AOGS).
- 2021 📖 **Graduate Award for Scientific Research**, Peking University.

Scholarships

- 2023 📖 **Peking University President's Scholarship**, Peking University.
- 2022 📖 **Chinese Government Scholarship**, China Scholarship Council.

Conference Organization

- 2026 📖 **Co-convenor**, Session ST29: Linking Plasma Dynamics in the Lower Solar Atmosphere to Magnetic Activity, AOGS Annual Meeting

Conference Presentations

Invited talks

- 2026 📖 The Contribution of Turbulence to Solar Wind Heating in the Inner Heliosphere, 2nd Plasma Astrophysics School and Workshop (PASW26).

Research Publications

Journal Articles (†, co-first author)

- 1 **C. Hou**, H. Yan, and S. Zhao, “Compressible turbulence as a source of particle beams and ion Bernstein waves in collisionless plasmas”, *The Astrophysical Journal* **1004**, 64 (2026).
- 2 **C. Hou**, J. He, A. Rouillard, et al., “Non-adiabatic entropy increase in the solar wind governed by velocity spikes”, *Nature Astronomy*, accepted (2026).
- 3 **C. Hou**, H. Yan, S. Zhao, and P. Pavaskar, “Energy cascade and damping in fast-mode compressible turbulence”, *The Astrophysical Journal Letters* **992**, L28 (2025).
- 4 **C. Hou**[†], B. Gannouni[†], A. P. Rouillard, J. He, and V. Réville, “Fine-scale activity driven by magnetic reconnection within coronal microjets”, *Astronomy & Astrophysics* **697**, A67 (2025).
- 5 **C. Hou**, J. He, D. Duan, Z. Wu, Y. Chen, D. Verscharen, A. P. Rouillard, H. Li, L. Yang, and S. D. Bale, “The origin of interplanetary switchbacks in reconnection at chromospheric network boundaries”, *Nature Astronomy* **8**, 1246–1256 (2024).
- 6 **C. Hou**, A. P. Rouillard, J. He, B. Gannouni, V. Réville, P. Louarn, A. Fedorov, L. Přech, C. J. Owen, D. Verscharen, R. D’Amicis, L. Sorriso-Valvo, N. Fargette, J. Coburn, V. Génot, J. M. Raines, R. Bruno, S. Livi, B. Lavraud, N. André, G. Fruit, R. Kieokaew, I. Plotnikov, E. Penou, A. Barthe, D. Kataria, M. Berthomier, F. Allegrini, V. Fortunato, G. Mele, and T. Horbury, “Connecting solar wind velocity spikes measured by solar orbiter and coronal brightenings observed by sdo”, *The Astrophysical Journal Letters* **968**, L28 (2024).
- 7 **C. Hou**[†], X. Zhu[†], R. Zhuo[†], J. He, D. Verscharen, and D. Duan, “Nature, generation, and dissipation of alfvénic kinks/switchbacks observed by Parker Solar Probe and Wind”, *The Astrophysical Journal* **950**, 157 (2023).
- 8 **C. Hou**, J. He, D. Duan, X. Zhu, W. Li, D. Verscharen, T. Liu, and T. Wang, “Efficient energy conversion through vortex arrays in the turbulent magnetosheath”, *The Astrophysical Journal* **946**, 13 (2023).
- 9 **C. Hou**, J. He, X. Zhu, and Y. Wang, “Contribution of magnetic reconnection events to energy dissipation in space plasma turbulence”, *The Astrophysical Journal* **908**, 237 (2021).
- 10 **C. Hou**, J. He, L. Zhang, Y. Wang, and D. Duan, “Dynamics of the charged particles released from a sun-grazing comet in the solar corona”, *Earth and Planetary Physics* **5**, 232–238 (2021).